

COORDINATION COMPOUNDS

1. Why is $[\text{NiCl}_4]^{-2}$ paramagnetic while $[\text{Ni}(\text{CN})_4]^{-2}$ is diamagnetic?
2. Why are low spin tetrahedral complexes are rarely observed?
3. When a coordination compound $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ is mixed with AgNO_3 , 2 moles of AgCl are precipitated per mole of the compound. Write the structural formula and IUPAC name of the complex.
4. For the complex $[\text{Fe}(\text{CN})_6]^{-4}$, write the hybridization, magnetic character and spin type of the complex.
5. State the hybridization and magnetic property of $[\text{Co}(\text{NH}_3)_6]^{+3}$ ion.
6. What is meant by crystal field splitting energy? On the basis of crystal field theory, write the electronic configuration for d^4 ion if (i) $\Delta_0 < P$ (ii) $\Delta_0 > P$
7. When a coordination compound $\text{CoCl}_3 \cdot 6\text{NH}_3$ is mixed with AgNO_3 , 3 moles of AgCl are precipitated per mole of the compound. Write the structural formula and IUPAC name of the complex.
8. For the complex $[\text{Fe}(\text{H}_2\text{O})_6]^{+3}$, write the hybridization, magnetic property and spin of the complex.
9. When a coordination compound $\text{PdCl}_3 \cdot 3\text{NH}_3$ is mixed with AgNO_3 , 2 moles of AgCl are precipitated per mole of the compound. Write the structural formula and IUPAC name of the complex.

10. For the complex $[\text{CoF}_6]^{-3}$, write the hybridization, magnetic property and spin of the complex.
11. Why is $[\text{Co(en)}_3]^{+3}$ more stable than the complex $[\text{CoF}_6]^{-3}$?
12. Which of the following is more stable?
 - i) $[\text{Co}(\text{NH}_3)_6]^{+3}$ or $[\text{Co}(\text{CO})_6]^{+3}$
 - ii) $[\text{Co}(\text{NH}_3)_6]^{+3}$ or $[\text{Co(en)}_3]^{+3}$
 - iii) $[\text{Co}(\text{H}_2\text{O})_6]^{+2}$ or $[\text{Co}(\text{H}_2\text{O})_6]^{+3}$
13. For the complex $[\text{Co}(\text{NH}_3)_6]^{+3}$, write the hybridization, magnetic property and spin of the complex.
14. $[\text{Fe}(\text{CN})_6]^{-4}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{+2}$ are of different colours in dilute solutions. Why?
15. Find the coordination number of the metal ion in $[\text{Co(ox)}_2\text{Cl}_2]^+$.
16. Using crystal field theory, draw energy level diagram, write electronic configuration (in terms of e_g and t_{2g}) for the central metal ion and determine the magnetic moment for the following:
 - i) $[\text{FeF}_6]^{-4}$
 - ii) $[\text{Co}(\text{NH}_3)_6]^{+3}$
17. On the basis of crystal field theory, write the electronic configuration for d^6 ion if (i) $\Delta_0 < P$ (ii) $\Delta_0 > P$
18. In the following complexes: $[\text{Fe}(\text{H}_2\text{O})_6]^{+2}$, $[\text{Co}(\text{CN})_6]^{-3}$ and $[\text{Ni}(\text{NH}_3)_6]^{+2}$ which complex will be
 - i) Diamagnetic and inner complex
 - ii) Paramagnetic and high spin complex

iii) Octahedral and outer complex

19. Draw the geometrical isomers of the complex $[\text{Pt}(\text{en})_2\text{Cl}_2]^{+2}$. Identify the isomer which is optically inactive.

20. A coordination compound $\text{CrCl}_3 \cdot 4\text{H}_2\text{O}$ precipitates silver chloride when treated with silver nitrate. The molar conductance of its solution corresponds to a total of two ions. Write the structural formula of the compound and name it. Identify the isomerism exhibited by this complex.

21. How many geometrical isomers are possible in $[\text{Al}(\text{C}_2\text{O}_4)_3]^{-3}$?

22. What type of isomerism is exhibited the following? Draw the structures of the possible isomers.

i) $[\text{Co}(\text{NH}_3)_2\text{Br}]\text{SO}_4$

ii) $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$

iii) $[\text{Co}(\text{NH}_3)_5\text{SCN}]^{+2}$

iv) $[\text{Co}(\text{en})]^{+3}$

v) $[\text{Pt}(\text{en})_2\text{Cl}_2]\text{SO}_4$

vi) $[\text{Co}(\text{en})_2\text{Br}_2]\text{SO}_4$

vii) $[\text{Fe}(\text{CN})_4(\text{NH}_3)_2]^{-}$

viii) $[\text{Cr}(\text{NH}_3)_2\text{Cl}_2(\text{en})]^{+}$

ix) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$

x) $[\text{Al}(\text{CO})_3(\text{NO}_2)_3]$